

Reflect Points Across Axes

Lesson 7-9

Name: Type your name

Date: Today's date

Class: Class period

My Notes

Level 2 Standard



TODAY'S OBJECTIVES

Content Objective: I can reflect points across the x-axis and y-axis on the coordinate plane.

Language Objective: I can explain my work using the words reflection, x-axis, y-axis, and symmetry.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Reflection

x-axis

y-axis

Symmetry

Integer

Perpendicular



Tap any word to see its meaning and a picture.

1

A flip of a point or shape over an axis to make a mirror image is a

.

2

The horizontal number line on the coordinate plane is the

.

3

The vertical number line on the coordinate plane is the

.

4

When two halves are mirror images of each other, the figure has

.

5

A whole number or its opposite, with no fraction part, is an

.

6

Two lines that meet at a 90° square corner are .

Write About the Math

How do reflections help the game designer create a balanced, symmetric level?

¿Cómo ayudan los reflejos a la diseñadora del juego a crear un nivel equilibrado y simétrico?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Reflection**
Reflexión

☐ **x-axis**
Eje x

☐ **y-axis**
Eje y

☐ **Symmetry**
Simetría

☐ **Integer**
Número entero

Model: *Reflecting over the y-axis changes the ____ coordinate because ____.* — Student states reflecting over the y-axis flips the sign of the x-coordinate while the y-coordinate stays the same, so (3,2) becomes (-3,2).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The enemy reflected over the y-axis is at ____ because ____.** The enemy reflected over the x-axis is at ____ because ____.
- **Reflections help game designers because ____.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.